Novel Data-Driven Nonlinear MPC for the Optimal Control of Air-Handling Units

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Abstract. Air-handling units (AHUs) have become indispensable parts of heating, ventilation, and air conditioning (HVAC) systems. AHUs are also significant energy consumers due to the function of their several actuators. Many recent works focus on improving the control techniques of AHUs to provide better indoor comfort with lower energy consumption. However, due to its inherent structure, it is complex to design an optimal and adaptive control for AHU that fulfills this mission in all operation conditions. Model predictive control (MPC), in this context, has been in focus in many contributions recently. However, designing a multi-input multi-output (MIMO) MPC for AHU optimal control is not a trivial task due to the difficulty of having a high-fidelity mathematical model. This study proposes and validates a data-driven nonlinear MPC with MIMO architecture. The proposed MPC is based on the sparse nonlinear dynamic of AHU built upon operation data of a real AHU installed in the KTH live-in lab. In contrast to the classical approaches, the proposed MPC adjusts simultaneously five different actuators to control the supply temperature. This article presents a simulation study for the performance of the proposed MPC framework under different control configurations.

Keywords: Air-handling unit control; data-driven model predictive control

1 INTRODUCTION

In recent years, data-driven approaches have become increasingly prevalent in the optimization of heating, ventilation, and air conditioning (HVAC) systems, particularly in areas such as fault detection, diagnosis, and energy management. These methods leverage real-time data to enhance operational efficiency and ensure more effective system performance. However, their application to optimal control, especially for air-handling units (AHUs), has been less explored. Traditional approaches often rely on simplistic models or controllers, such as proportional-integral-derivative (PID) controllers, which fail to capture the full complexity of AHU systems. Recent advancements have seen the introduction of more sophisticated techniques, which aim to address some of the limitations of these conventional methods. However, challenges remain in creating models that accurately represent AHU dynamics and in developing control strategies that can efficiently manage system energy consumption while ensuring comfort.

Data-driven approaches have been extensively utilized in HVAC systems for various applications, including fault detection and diagnosis [1] and energy management [2]. However, their application to optimal control, particularly for air-handling units (AHUs), remains relatively limited. Ghawash et al. [3] proposed and simulated a nonlinear optimizer for AHUs to regulate zone temperatures using simple polynomial models calibrated with measurement data. While the simulation demonstrated energy-efficient operation, the models used were simplistic and failed to capture all internal AHU states. In [4], the authors introduced and validated a deep reinforcement learning (DRL) approach for optimal AHU control. The DRL agents achieved energy savings of 27%–30% compared to actual energy consumption while keeping the predicted percentage of discomfort at 10%. This study relied on a long-short-term memory (LSTM) model trained on real operational data to approximate HVAC performance. However, a significant limitation was the LSTM's inability to account for the various internal parameters of the AHU. Asvadi-Kermani et al. [5] developed predictive controllers for AHUs based on a state-space model estimated using a dynamic recurrent neural network (RNN). Their approach resulted in a 69.9% reduction in energy consumption compared to the real system's dataset. Nonetheless, oversimplifying the AHU model structure within the control framework was identified as a key limitation.

This article introduces an innovative data-driven framework for model-based control of AHU supply temperature. In contrast to conventional methods that rely solely on PID controllers to regulate the supply temperature through the heating battery, this approach employs multiple AHU actuators working together to optimize energy consumption during the regulation process. By utilizing data from the building management system (BMS), a nonlinear multi-input multi-output (MIMO) model of the AHU is developed based on sparse identification of nonlinear dynamics (SINDy), enabling the application of nonlinear model predictive control (NMPC) for more efficient and adaptive regulation.

2 METHOD

In this section, we present the methodology for developing NMPC based on SINDy technique (SINDy-NMPC) designed to drive AHU as an optimal regulator instead of the classical PID. The proposed technique aims to regulate the AHU air supply temperature to the setpoint, sent by a BMS controller, with less control energy. This can be achieved through the real-time adjustment of the recycled/intake air mix thanks to the optimal coordination of multiple controllable devices which are the heating batteries, supply and exhaust fans, and rotating heat exchangers as illustrated in Fig. 1. Supply and exhaust dampers can be included as well in this control framework, but they are not considered in this study.

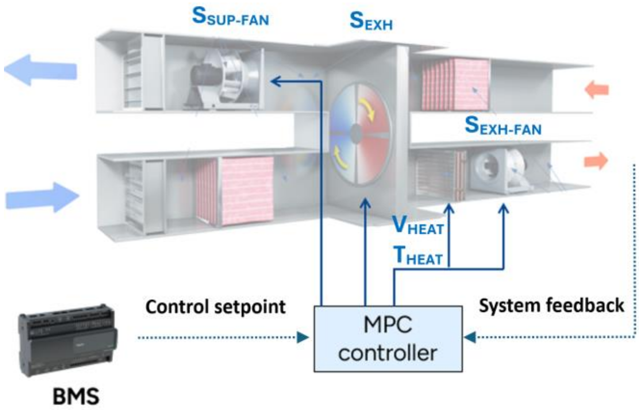

Fig. 1. Diagram of the AHU supply temperature control using the proposed MPC, where $S_{SUP\text{-}FAN}$ and $S_{HEX\text{-}FAN}$ are respectively the supply and exhaust fan speeds, S_{EXH} is the rotating heat exchanger speed, V_{HEAT} and T_{HEAT} are respectively the water supply and opening valve rate of the heating battery.

2.1 AHU modeling using sparse identification

Analyzing the governing equations of AHUs, a multivariable nonlinear dynamic system, is challenging due to the complex interactions between various states. In situations, sparse identification using SINDy proves to be a powerful technique as it combines the flexibility of nonlinear modeling with the simplicity and interpretability of a linear representation. By enforcing sparsity, SINDy avoids overfitting and ensures that the resulting model remains interpretable and free from unnecessary complexity. Eq (1) shows the general SINDy mathematical representation of a nonlinear system, where x is the system state vector; u is the input vector, Θ is the nonlinear candidate functions and ξ is the regression coefficient matrix. This later determines the active terms in Θ.

$$\dot{x} = \Theta(x(t), u(t)).\xi \qquad (1)$$

To find out the optimal coefficients of ξ, a regression problem is to be formulated using data with equal time steps (snapshots), where the objective is to minimize the mathematical expression shown in (2).

$$\min_{\xi} \sum_{j=1}^{N_{snap}-1} \left\| \Theta(x(j), u(j))\xi - x(j+1) \right\|_2^2 + \sigma \|\xi\| \qquad (2)$$

Where j is the data snapshot index; σ is the regularization coefficient to incorporate sparsity; and N_{SNAP} is the number of total data snapshots.

2.2 Nonlinear MPC formulation

As stated earlier, the main goal of the proposed MPC formulation is to find out the optimal operation coordination of several AHU actuators to stabilize the air supply temperature around a predefined setpoint with less control energy. To do so, the MPC is in charge of cyclically optimizing the following objective function:

$$\min \sum_{i=1}^{N} (y_D - y)^T Q (y_D - y) + (u - u_0)^T R (u - u_0) \tag{3}$$

Subject to:

$$\begin{aligned} & y = \Theta(x, u).\xi \\ & u_{\min} < u < u_{\max} \end{aligned} \tag{4}$$

Where y is the vector of measured states; y_D is the vector of the desired setpoints for the measured variables (in our case, only air supply temperature); $u_{\min}$ and $u_{\max}$ are respectively the minimum and maximum limits for input variables; Q and R are respectively the state and control weighting matrices; N is the control horizon.

2.3 Use-case validation

To validate the proposed SINDy-NMPC framework under realistic operational conditions, a use-case study was performed using data from an AHU installed at the KTH Live-In Lab (see Figs. 2). This AHU is designed to meet the heating demands of four student apartments. Three years of historical data from the lab's BMS database were used to construct the SINDy model. Nonlinear candidate functions were represented as second-degree linear combinations of the state and input variables. The resulting model was integrated into an NMPC framework and tested through digital simulations, as real-world implementation on the AHU was not feasible at this stage.

For this validation, the following state and input vectors were selected:

$x = [T_{SUP}, F_{SUP}, T_{REC}, T_{EXH}, F_{EXH}]$ and

$u = [V_{HEAT}, T_{HEAT}, S_{SUP-FAN}, S_{HEX}, S_{EXH-FAN}]$

Where T_{SUP} is the supply air temperature [°C]; F_{SUP} is the supply airflow [l/min]; T_{REC} is the recycled air temperature [°C]; T_{EXH} is the exhaust air temperature [°C]; F_{EXH} is the exhaust airflow [l/min]; V_{HEAT} is the battery heating rate (valve opening rate) [%]; T_{HEAT} the battery heating temperature [°C]; $S_{SUP-FAN}$ is the supply fan speed [%]; S_{HEX} is the rotating heat exchanger speed [%]; $S_{EXH-FAN}$ is the exhaust fan speed [%];

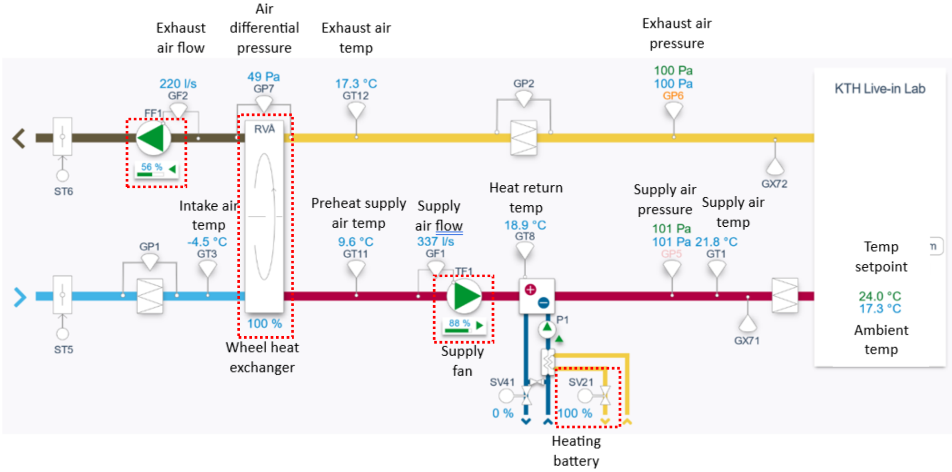

Fig. 2. SCADA view of AHU of the KTH live-in lab highlighting the actuators considered for SINDy-NMPC

As the proposed SINDy-NMPC framework will be evaluated solely through simulations, a high-fidelity deep learning model for the AHU, known as a neural state-space (NSS) model, was developed. Unlike SINDy, NSS requires a substantial amount of data for training as it captures system nonlinearities within a large multi-layer neural network (see Fig. 3). While it functions as a black-box model, NSS offers higher prediction accuracy, making it ideal for evaluating the SINDy-NMPC framework. The parameters used for NSS training are detailed in Table 1. The NSS system state incorporates the input vector *u* while the output *y* represents the AHU supply temperature.

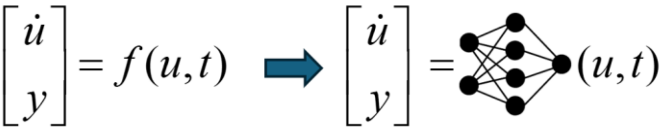

Fig. 3. Concept of NSS where *u* is the model input, *y* is the output and *t* time

Table 1. Training parameters of the AHU neural state-space model

Training parameters	Value	Training parameters	Value
Solver	Adam	Number of neurons in the input layer	5
Maximum epochs	200	Number of neurons in the hidden layers	256
Minimum batch size	999	Number of neurons in the output layer	1
Number of hidden layers	3		

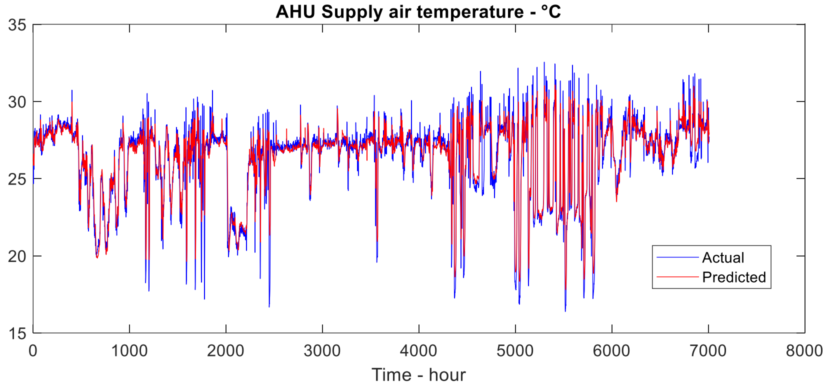

Fig. 4. Validation data of the AHU supply temperature prediction using NSS

As can be seen on Fig. 4, using one-year hourly data, NSS could capture the AHU nonlinearity and give acceptable supply temperature prediction performance.

Fig. 5 represents the general overview of the validation procedure, where SINDy has been used to predict the NSS response and use sequential quadratic programming (SQP) to find out the optimal control inputs that optimize Eq (3) at each time step. Based on Q and R selection, one can choose how MPC drives the actuators to provide the desired supply temperature setpoint with less control energy.

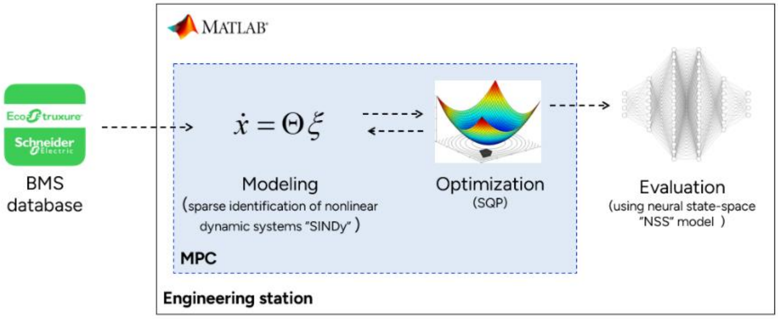

Fig. 5. Proposed simulation framework for SINDy-NMPC

3 RESULTS AND DISCUSSION

In this section, a simulation study will be carried out to assess the proposed SINDy-NMPC on the developed AHU NSS model using real data queried from the KTH live-in lab database. The models and algorithms are implemented in MATLAB and run on Intel(R) Core(TM) i7-1185G7, 4 cores CPU. First, we evaluate the SINDy prediction performance using real AHU data. Then, we simulate the SINDy-NMPC performance for AHU supply temperature regulation for different control configurations.

3.1 AHU states prediction using SINDy

Fig. 6 represents the one-hour ahead prediction performance of SINDy, which is also the control horizon that will be chosen for NMPC. The model was trained by fetching and using historical data of the AHU operation from the BMS database. The nonlinear candidate functions are a second-degree polynomial that gathers all possible combinations of system states and inputs. We see clearly how the proposed sparse identification could accurately predict all AHU states using one unique MIMO model with a normalized global mean squared error (MSE) of 0.23. Going for a three-hour prediction horizon led to a normalized MSE of 0.62.

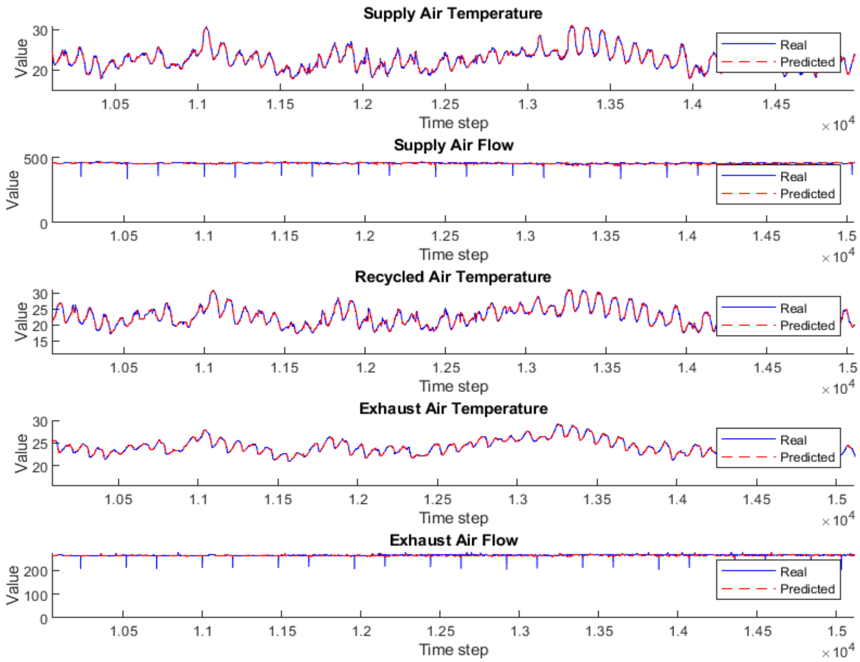

Fig. 6. One-hour ahead prediction of different AHU states using SINDy (here time step is 15 min)

3.2 SINDy-NMPC performance

The NMPC has been set using the parameters listed in Table 2 for all simulation scenarios. However, different objective functions with different weighting matrices Q and R have been simulated within the upcoming cases.

Table 2. MPC settings

Training parameters	Value	Training parameters	Value
Time step	15 minutes	Solver	SQP
Prediction horizon	1 hour	Control horizon	1 hour

Case 1: Baseline case $Q = 1$ and $R = 1e^{-5}[1,1,1,1,1]$

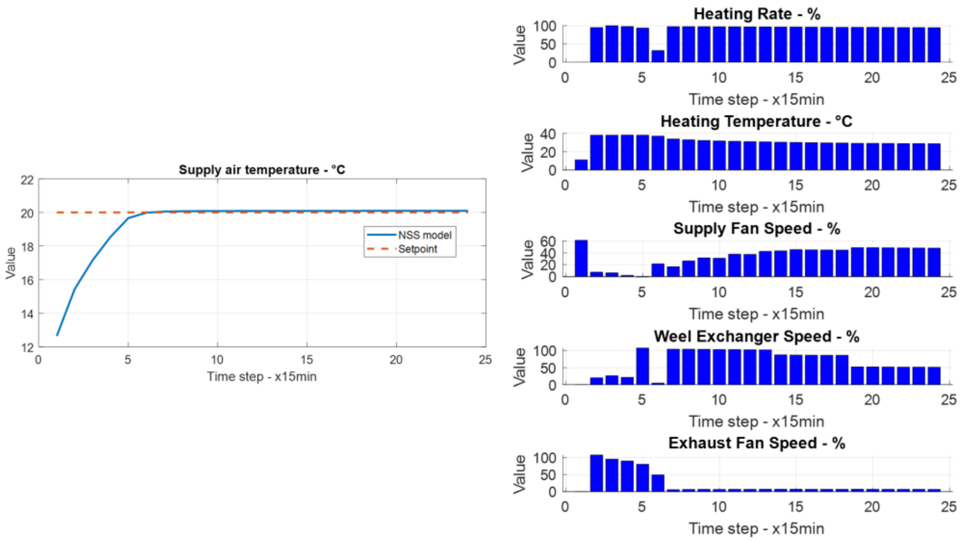

Fig. 7. SINDy-NMPC performance for six-hour operation simulation: baseline scenario

Case 2: penalty on the energy cost from the heating battery case $Q = 1$ and $R = [1e^{2}, 1e^{2}, 1, 1, 1]$.

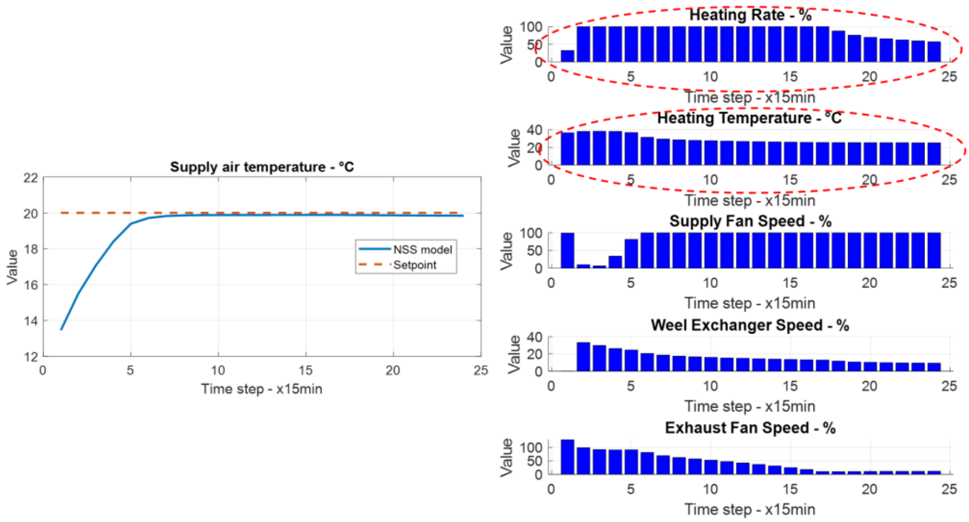

Fig. 8. SINDy-NMPC performance for six-hour operation simulation: penalty on the heating battery

Case 3: penalty on the energy cost from the fans $Q = 1$ and $R = [1,1,1e^3,1,1e^3]$

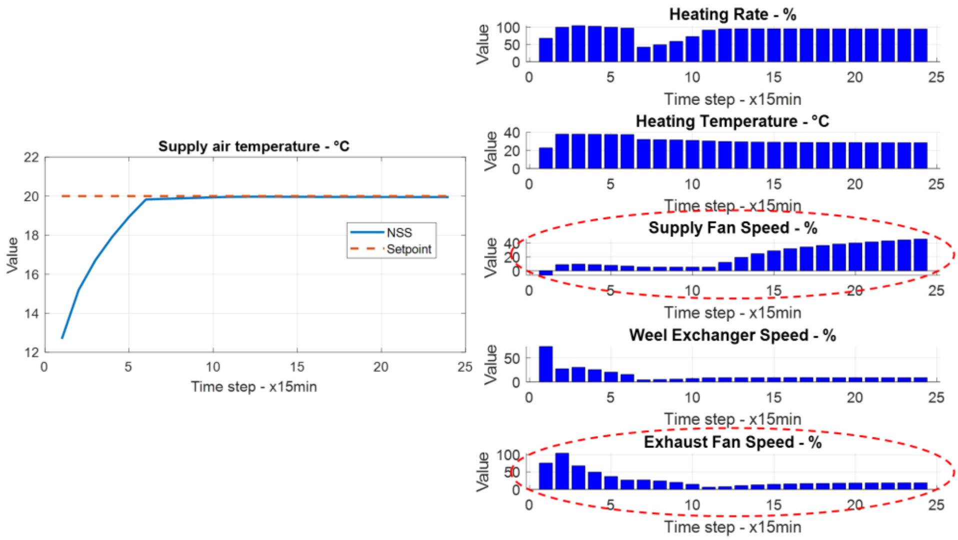

Fig. 9. SINDy-NMPC performance for six-hour operation simulation: penalty on the fans' operation

In the baseline scenario, the Q and R matrices in the NMPC cost function were chosen equally for all elements when regulating the air supply temperature, which gave this balanced operation of different AHU actuators shown in Fig. 7. In Case 2, R is modified so the coefficients corresponding to the AHU heating battery have been accurately maximized, which explains its relatively reduced operation in terms of heating temperature and rate in (%) (see Fig. 8). This is a typical case when we have an electrically driven heating system with a high electricity price. In Case 3, where there is a need to monitor the fans' operation due to their electricity consumption, here R matrix is updated, this time by accurately maximizing the coefficient corresponding to the supply and exhaust fans. Fig. 9 shows how the operation rate of the two fans has been moderated while maintaining the same supply temperature regulation performance. The proposed NMPC aims to find out the optimal mix between intake and recycled air at each operation condition through real-time adjustment of the AHU actuators. Based on the objective function formulation, this will enable effective setpoint tracing of air supply temperature with minimized control energy. For a more fair and effective comparison, the three NMPC configurations shall be compared with the current regulation system implemented in the KTH live-in lab, which is a PID-controlled heating battery. To do so, future work will incorporate a simulation study for the PID use case with the same operation conditions.

4 CONCLUSIONS

This paper presents a simulation study developed and validated for the optimal control of supply air temperature in air-handling units (AHUs). The proposed method utilizes a data-driven nonlinear model predictive control (MPC) framework built on a sparse identification model. The simulation results demonstrate the accuracy of sparse identification in capturing system complexity and nonlinearities within an overall linear representation, enabling the implementation of optimal control. Additionally, the approach highlights the effectiveness of MPC in achieving robust and fast setpoint tracking by simultaneously coordinating multiple actuators, offering enhanced control flexibility for optimizing energy use.

Future work will expand the control framework to include supply airflow, allowing the total supplied heat to be regulated. Furthermore, the building's indoor environment will be incorporated by integrating the building's thermal mass into the overall model. This integration will extend MPC's flexibility to consider the indoor environment, thereby enhancing the overall optimal control strategy.

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